

Havelock Lagrangian Eigenvalues: First-Principles Derivation

Sandbox for Paper I, Item #1

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Main result

Theorem 1 (Havelock Lagrangian eigenvalues). *On the unit ring $z_k = e^{2\pi i k/N}$, $k = 0, \dots, N-1$, of unit-circulation point vortices with energy $H = -\sum_{j < k} \ln |z_j - z_k|$, the Lagrangian Hessian $\nabla^2 \mathcal{L} = \nabla^2 H - 2\mu_L I$ with Lagrange multiplier $\mu_L = -(N-1)/4$ is block-diagonal in the $\mathbb{Z}_N$ Fourier basis. At each mode $m \in \{1, \dots, N-1\}$ the 2×2 block in the normalized (radial, tangential) basis has vanishing off-diagonal entry and eigenvalues*

$$\lambda_m^+ = (N-1) - \frac{m(N-m)}{2}, \quad \lambda_m^- = \frac{m(N-m)}{2}.$$

In particular, $\lambda_m^+ + \lambda_m^- = N-1$ is mode-independent.

Proof. By Lemma 4 (off-diagonal vanishing), the mode- m block is diagonal, so its eigenvalues are the two diagonal entries. The radial-radial entry is computed in §3, equation (13): $(\nabla^2 H)_{rr}^{(m)} = (N-1 - m(N-m))/2$, built from the Havelock identity (Lemma 2) and the cosecant-squared sum (Lemma 3, Corollary 1). The tangential-tangential entry is computed in §4, equation (17): $(\nabla^2 H)_{tt}^{(m)} = (m(N-m) - (N-1))/2 = C(1) - C(m)$. The Lagrange shift $(N-1)/2$ applied to each diagonal entry (§6) yields (19). The trace identity (Lemma 5) then follows by direct addition. $\square$

The remainder of this document gives the derivation in full, each step verified numerically against the definition-based oracle in `numerical.check.py` (symbolic identities to 30 decimal digits; oracle matches the full-diagonalization Hessian to 10^{-2} , finite-difference precision) for $N \in \{3, \dots, 12\}$ and $m \in \{1, \dots, N-1\}$.

1 Setup

Let $z_k = e^{i\theta_k}$, $\theta_k = 2\pi k/N$, $k = 0, \dots, N-1$, be the regular N -gon of unit-circulation point vortices on the unit circle. The Thomson energy is

$$H = -\sum_{j < k} \ln |z_j - z_k|. \tag{1}$$

We perturb each vortex in its *local* (radial, tangential) frame:

$$z_k \longmapsto z_k(1 + \delta r_k) + i z_k \delta t_k, \tag{2}$$

with $\delta r_k, \delta t_k \in \mathbb{R}$ small.

$\mathbb{Z}_N$ -equivariance. The cyclic action $\phi : k \mapsto k+1 \pmod{N}$ on vortex labels is a symmetry of the equilibrium ring and of H : relabeling vortices permutes the pair sum (1) invariantly. Under ϕ the perturbation data transforms as $(\delta r_k, \delta t_k) \mapsto (\delta r_{k-1}, \delta t_{k-1})$, because the *local* radial and tangential directions at vertex $k+1$ are the images of the corresponding directions at vertex k under the Euclidean rotation by $2\pi/N$ that maps $z_k \mapsto z_{k+1}$. The Hessian $\nabla^2 H$ is therefore $\mathbb{Z}_N$ -equivariant on the $2N$ -dimensional perturbation space $\mathbb{R}^{2N} \cong \mathbb{R}^N \oplus \mathbb{R}^N$ (radial $\oplus$ tangential). By the standard finite-group representation theory, $\nabla^2 H$ block-diagonalizes in the $\mathbb{Z}_N$ isotypic components, indexed by $m \in \{0, 1, \dots, N-1\}$.

Real Fourier modes. The $\mathbb{Z}_N$ -irreducibles over $\mathbb{R}$ are two copies of the trivial rep at $m = 0$, two copies of the sign rep at $m = N/2$ (even N only), and for $m \in \{1, \dots, \lfloor (N-1)/2 \rfloor\}$ a paired complex representation $(m, N-m)$ of real dimension 4 per pair, spanned by

$$\delta r_k = a_c \cos \frac{2\pi m k}{N} + a_s \sin \frac{2\pi m k}{N}, \quad \delta t_k = b_c \cos \frac{2\pi m k}{N} + b_s \sin \frac{2\pi m k}{N}. \quad (3)$$

So at each $m \in \{1, \dots, N-1\}$ with $m \neq N/2$, the Hessian restricts to a 4×4 block in the (a_c, a_s, b_c, b_s) basis.

Lemma 1 (Cos-sin equivalence). *At every $m \in \{1, \dots, N-1\}$ with $m \neq N/2$, the 4×4 Hessian block decomposes as two identical 2×2 blocks, one on (a_c, b_c) and one on (a_s, b_s) , and it suffices to compute the cosine sub-block.*

Proof. The $\text{SO}(2)$ rotation $R_\beta : z_k \mapsto e^{i\beta} z_k$ (rigid rotation of the ring) is a symmetry of H and preserves the equilibrium for every β . It acts on mode- m amplitudes by $(a_c + ia_s) \mapsto e^{-i\beta m} (a_c + ia_s)$ (and similarly on (b_c, b_s)), because the cos/sin mode at m is the real/imaginary part of $e^{2\pi i m k/N}$, and a rotation by β shifts the vertex angles by β . In particular, the rotation $R_{\pi/(2m)}$ maps $(a_c, a_s) \rightarrow (a_s, -a_c)$ and $(b_c, b_s) \rightarrow (b_s, -b_c)$. Since $R_{\pi/(2m)}$ is a symmetry of $\nabla^2 H$, the 4×4 block commutes with this swap, so its (a_c, b_c) and (a_s, b_s) 2×2 sub-blocks are equal, and any (a_c, b_s) or (a_s, b_c) cross entries are zero. Palindromic symmetry $m \leftrightarrow N-m$ is the complex conjugate of this statement. $\square$

The mode- m cosine sub-block — which is what we compute below — is therefore sufficient. In the notation we use throughout,

$$\delta r_k = a \cos(2\pi m k/N), \quad \delta t_k = b \cos(2\pi m k/N), \quad (4)$$

parametrized by (a, b) , which yields the 2×2 block computed in §3–§6.

The case $m = N/2$ (even N). When N is even and $m = N/2$, the paired representation degenerates to a single real dimension per vertex direction ($\sin(2\pi \cdot N/2 \cdot k/N) = \sin(\pi k) = 0$, so only the cosine mode exists). In this case $\alpha_k := \cos(\pi k) = (-1)^k$, so $\sum_k \alpha_k^2 = N$ (not $N/2$) and the normalization of the mode vectors doubles. However, the elementary Fourier sums also double correspondingly ($\sum_j \alpha_j \alpha_{j+p} = N(-1)^p = N \cos(\pi p)$, twice the generic formula $(N/2) \cos(2\pi m p/N)$ evaluated at $m = N/2$), and the two factors of two cancel in the normalized Hessian entry. The final eigenvalue formula thus extends uniformly to $m = N/2$; we record this explicitly as Remark 1 after the main derivation.

2 Two auxiliary identities

Two closed-form trigonometric sums enter the block computation and are recorded here. Both are verified to 30 decimal digits in `numerical_check.py` (`check_havelock_identity_rr`, `check_cosecant_squared_sum`).

Lemma 2 (Havelock identity). *For all integers $N \geq 2$ and $m \in \{1, \dots, N-1\}$,*

$$S(m) := \sum_{p=1}^{N-1} \frac{1 - \cos(2\pi mp/N)}{4 \sin^2(\pi p/N)} = \frac{m(N-m)}{2}. \quad (5)$$

Proof. Using $1 - \cos(2\pi mp/N) = 2 \sin^2(\pi mp/N)$,

$$S(m) = \sum_{p=1}^{N-1} \frac{\sin^2(\pi mp/N)}{2 \sin^2(\pi p/N)}. \quad (6)$$

We derive the identity $\sum_{p=1}^{N-1} \sin^2(\pi mp/N) / \sin^2(\pi p/N) = m(N-m)$ from first principles. Write the telescoping geometric sum

$$e^{-i(m-1)\pi p/N} + e^{-i(m-3)\pi p/N} + \dots + e^{i(m-1)\pi p/N} = \sum_{k=0}^{m-1} e^{i(2k-m+1)\pi p/N} = \frac{e^{im\pi p/N} - e^{-im\pi p/N}}{e^{i\pi p/N} - e^{-i\pi p/N}} = \frac{\sin(\pi mp/N)}{\sin(\pi p/N)}.$$

Squaring the modulus,

$$\frac{\sin^2(\pi mp/N)}{\sin^2(\pi p/N)} = \sum_{k=0}^{m-1} \sum_{\ell=0}^{m-1} e^{i(2k-2\ell)\pi p/N} = \sum_{k=0}^{m-1} \sum_{\ell=0}^{m-1} e^{2\pi i(k-\ell)p/N}.$$

Summing over $p = 1, \dots, N-1$ and using the orthogonality $\sum_{p=0}^{N-1} e^{2\pi iqp/N} = N \mathbf{1}\{q \equiv 0 \pmod{N}\}$ (hence $\sum_{p=1}^{N-1} e^{2\pi iqp/N} = -1 + N \mathbf{1}\{q \equiv 0\}$),

$$\sum_{p=1}^{N-1} \frac{\sin^2(\pi mp/N)}{\sin^2(\pi p/N)} = \sum_{k,\ell=0}^{m-1} (N \mathbf{1}\{k \equiv \ell \pmod{N}\} - 1) = N \cdot m - m^2 = m(N-m),$$

since for $k, \ell \in \{0, \dots, m-1\}$ with $m \leq N-1$, the condition $k \equiv \ell \pmod{N}$ is equivalent to $k = \ell$, giving m diagonal contributions each of value N , minus m^2 constant -1 's. Dividing by 2 gives $S(m) = m(N-m)/2$. $\square$

Lemma 3 (Cosecant-squared sum). *For $N \geq 2$,*

$$\sum_{p=1}^{N-1} \frac{1}{4 \sin^2(\pi p/N)} = \frac{N^2 - 1}{12}. \quad (7)$$

Proof. Sum Lemma 2 over $m = 1, \dots, N-1$:

$$\sum_{m=1}^{N-1} S(m) = \sum_{m=1}^{N-1} \frac{m(N-m)}{2} = \frac{1}{2} \left[N \cdot \frac{N(N-1)}{2} - \frac{(N-1)N(2N-1)}{6} \right] = \frac{(N-1)N(N+1)}{12}.$$

On the other hand, interchanging summation order in the definition of $S(m)$ and using $\sum_{m=1}^{N-1} (1 - \cos(2\pi mp/N)) = (N-1) - (-1) = N$ for each fixed $p \in \{1, \dots, N-1\}$ (since $\sum_{m=0}^{N-1} \cos(2\pi mp/N) = 0$ and the $m=0$ term contributes 1),

$$\sum_{m=1}^{N-1} S(m) = \sum_{p=1}^{N-1} \frac{\sum_{m=1}^{N-1} (1 - \cos(2\pi mp/N))}{4 \sin^2(\pi p/N)} = N \sum_{p=1}^{N-1} \frac{1}{4 \sin^2(\pi p/N)}.$$

Equating the two expressions and dividing by N gives $\sum_{p=1}^{N-1} 1/(4 \sin^2(\pi p/N)) = (N^2 - 1)/12$. $\square$

Corollary 1 (Cosine-weighted inverse-sine sum). *For $N \geq 2$ and any integer k ,*

$$C(k) := \sum_{p=1}^{N-1} \frac{\cos(2\pi kp/N)}{4 \sin^2(\pi p/N)} = \frac{N^2 - 1}{12} - \frac{k(N - k)}{2} \quad \text{for } 0 \leq k \leq N. \quad (8)$$

The expression is periodic in k modulo N , i.e. $C(k + N) = C(k)$.

Proof. Subtract (5) from (7): $C(k) = (N^2 - 1)/12 - S(k) = (N^2 - 1)/12 - k(N - k)/2$. Periodicity is immediate from the integrand. $\square$

3 Radial-radial block

Under a purely radial mode- m perturbation $\delta r_k = a \alpha_k$ with $\alpha_k = \cos(2\pi mk/N)$ (and $b = 0$ in (3)), the squared pair distance expands as

$$|z_j(a) - z_k(a)|^2 = |z_j - z_k|^2 [1 + a(\alpha_j + \alpha_k)] + a^2 Q_{jk} + O(a^3), \quad (9)$$

with

$$Q_{jk} := \alpha_j^2 + \alpha_k^2 - 2\alpha_j \alpha_k \cos(\theta_j - \theta_k), \quad |z_j - z_k|^2 = 4 \sin^2(\pi(j - k)/N).$$

(To see this, write $z_k(a) = z_k(1 + a\alpha_k)$, compute $|z_j(a) - z_k(a)|^2 = |z_j - z_k|^2 + 2a \operatorname{Re}[(\overline{z_j - z_k})(\alpha_j z_j - \alpha_k z_k)] + a^2 |\alpha_j z_j - \alpha_k z_k|^2$, and simplify the real part using $\operatorname{Re}[(e^{-i\theta_j} - e^{-i\theta_k})(\alpha_j e^{i\theta_j} - \alpha_k e^{i\theta_k})] = (\alpha_j + \alpha_k)(1 - \cos(\theta_j - \theta_k))$.)

Expanding $-\ln |z_j(a) - z_k(a)| = -\frac{1}{2} \ln(|z_j(a) - z_k(a)|^2)$ to second order in a via $\ln(1 + x) = x - x^2/2 + O(x^3)$,

$$-\ln |z_j(a) - z_k(a)| = -\ln |z_j - z_k| - \frac{a}{2}(\alpha_j + \alpha_k) + a^2 \left[\frac{(\alpha_j + \alpha_k)^2}{4} - \frac{Q_{jk}}{2|z_j - z_k|^2} \right] + O(a^3). \quad (10)$$

The linear term, summed over pairs $j < k$, gives $-(a/2)(N-1) \sum_k \alpha_k = 0$ since $\sum_k \cos(2\pi mk/N) = 0$ for $m \in \{1, \dots, N-1\}$ (equilibrium).

Write β_m for the coefficient of a^2 in $H(a) - H(0)$:

$$\beta_m = \sum_{j < k} \left[\frac{(\alpha_j + \alpha_k)^2}{4} - \frac{Q_{jk}}{2|z_j - z_k|^2} \right].$$

The unit-normalized radial Fourier mode vector $\hat{e}_r$ (in the $2N$ -dimensional Cartesian basis) has squared norm $\|\hat{e}_r\|^2 = \sum_k \alpha_k^2 = N/2$ for $m \in \{1, \dots, N-1\}$ with $m \neq N/2$; for even N at $m = N/2$ the norm is N — see Remark 1 below for the explicit parallel computation that yields the same final formula. The Hessian entry in this normalized direction is

$$(\nabla^2 H)_{rr}^{(m)} = \frac{H''(0)}{\|\hat{e}_r\|^2} = \frac{2\beta_m}{N/2} = \frac{4\beta_m}{N} = \frac{2}{N} \sum_{j \neq k} \left[\frac{(\alpha_j + \alpha_k)^2}{4} - \frac{Q_{jk}}{2|z_j - z_k|^2} \right], \quad (11)$$

where we used $H''(0) = 2\beta_m$ (for a Taylor expansion $H(a) - H(0) = \beta_m a^2 + O(a^3)$) and $\sum_{j < k} = \frac{1}{2} \sum_{j \neq k}$ (the summand is symmetric in $j \leftrightarrow k$).

Parametrize pairs by $p = k - j \pmod{N}$ with $p \in \{1, \dots, N-1\}$ and apply the N -fold cyclic symmetry. Two elementary sums are needed:

$$\sum_{j=0}^{N-1} \alpha_j^2 = \frac{N}{2}, \quad \sum_{j=0}^{N-1} \alpha_j \alpha_{j+p} = \frac{N}{2} \cos(2\pi mp/N).$$

From these,

$$\sum_{j=0}^{N-1} (\alpha_j + \alpha_{j+p})^2 = N(1 + \cos(2\pi mp/N)), \quad \sum_{j=0}^{N-1} Q_{j,j+p} = N(1 - \cos(2\pi mp/N) \cos(2\pi p/N)).$$

Substituting into (11),

$$(\nabla^2 H)_{rr}^{(m)} = \sum_{p=1}^{N-1} \frac{1 + \cos(2\pi mp/N)}{2} - \sum_{p=1}^{N-1} \frac{1 - \cos(2\pi mp/N) \cos(2\pi p/N)}{4 \sin^2(\pi p/N)}. \quad (12)$$

The first sum is elementary: $\sum_{p=1}^{N-1} (1 + \cos(2\pi mp/N))/2 = (N-1)/2 - 1/2 = (N-2)/2$, since $\sum_{p=1}^{N-1} \cos(2\pi mp/N) = -1$ for $m \in \{1, \dots, N-1\}$.

The second sum reduces via the product-to-sum identity $\cos A \cos B = \frac{1}{2}[\cos(A-B) + \cos(A+B)]$:

$$1 - \cos(2\pi mp/N) \cos(2\pi p/N) = 1 - \frac{1}{2} \cos(2\pi(m-1)p/N) - \frac{1}{2} \cos(2\pi(m+1)p/N).$$

Applying Lemma 3 and Corollary 1,

$$\begin{aligned} \sum_{p=1}^{N-1} \frac{1 - \cos(2\pi mp/N) \cos(2\pi p/N)}{4 \sin^2(\pi p/N)} &= \frac{N^2 - 1}{12} - \frac{1}{2} \left[\frac{N^2 - 1}{12} - \frac{(m-1)(N-m+1)}{2} \right] \\ &\quad - \frac{1}{2} \left[\frac{N^2 - 1}{12} - \frac{(m+1)(N-m-1)}{2} \right] \\ &= \frac{(m-1)(N-m+1) + (m+1)(N-m-1)}{4} = \frac{m(N-m) - 1}{2}. \end{aligned}$$

Combining,

$$(\nabla^2 H)_{rr}^{(m)} = \frac{N-2}{2} - \frac{m(N-m) - 1}{2} = \frac{N-1-m(N-m)}{2}. \quad (13)$$

This is the radial-radial block eigenvalue of the *unshifted* energy Hessian $\nabla^2 H$. The Lagrange shift is applied in §6.

4 Tangential-tangential block

Under a purely tangential mode- m perturbation $\delta t_k = b \alpha_k$ with $\alpha_k = \cos(2\pi mk/N)$ (and $a = 0$ in (3)), the perturbation is $\delta z_k = i z_k b \alpha_k$, so $z_k(b) = z_k(1 + i b \alpha_k)$. A direct computation yields

$$|z_j(b) - z_k(b)|^2 = |z_j - z_k|^2 + 2b(\alpha_j - \alpha_k) \sin(\theta_j - \theta_k) + b^2 Q_{jk} + O(b^3), \quad (14)$$

with the same Q_{jk} as in (9). With $A = z_j - z_k$ and $B = \alpha_j z_j - \alpha_k z_k$,

$$|z_j(b) - z_k(b)|^2 = |A + i b B|^2 = |A|^2 - 2b \operatorname{Im}(\bar{A} B) + b^2 |B|^2,$$

and $\operatorname{Im}(\bar{A} B) = \operatorname{Im}[(z_j - z_k)^*(\alpha_j z_j - \alpha_k z_k)] = -(\alpha_j - \alpha_k) \sin(\theta_j - \theta_k)$ (from $\operatorname{Im}[\alpha_j + \alpha_k - \alpha_k e^{i(\theta_k - \theta_j)} - \alpha_j e^{i(\theta_j - \theta_k)}]$). The sign in (14) follows: $-2b \cdot [-(\alpha_j - \alpha_k) \sin(\theta_j - \theta_k)] = +2b(\alpha_j - \alpha_k) \sin(\theta_j - \theta_k)$.

Expanding $-\frac{1}{2} \ln |z_j(b) - z_k(b)|^2$ to second order in b (with $M_{jk} = 2(\alpha_j - \alpha_k) \sin(\theta_j - \theta_k)$ and $U_{jk} = |z_j - z_k|^2$),

$$-\ln |z_j(b) - z_k(b)| = -\ln |z_j - z_k| - \frac{b M_{jk}}{2 U_{jk}} + b^2 \left[\frac{M_{jk}^2}{4 U_{jk}^2} - \frac{Q_{jk}}{2 U_{jk}} \right] + O(b^3).$$

The linear term, summed over pairs, vanishes: setting $p = k - j$ and pairing p with $N - p$, the factor $\sin(\theta_j - \theta_k) = -\sin(2\pi p/N)$ is odd in p while $(\alpha_j - \alpha_k)/U_{jk}$ is even in p , so $\sum_{j < k} M_{jk}/U_{jk} = 0$ term by term.

With $p = k - j \pmod{N}$ and $\theta_j - \theta_k = -2\pi p/N$,

$$U_{jk} = 4 \sin^2(\pi p/N), \quad M_{jk}^2 = 4(\alpha_j - \alpha_k)^2 \sin^2(2\pi p/N) = 4(\alpha_j - \alpha_k)^2 \cdot U_{jk} \cos^2(\pi p/N),$$

using $\sin^2(2\pi p/N) = 4 \sin^2(\pi p/N) \cos^2(\pi p/N)$. Therefore $M_{jk}^2/(4U_{jk}^2) = (\alpha_j - \alpha_k)^2 \cos^2(\pi p/N)/U_{jk}$.

Proceeding as in §3, applying $\sum_j (\alpha_j - \alpha_{j+p})^2 = N(1 - \cos(2\pi mp/N))$ and $\sum_j Q_{j,j+p} = N(1 - \cos(2\pi mp/N) \cos(2\pi p/N))$, the tangential-tangential block entry is

$$(\nabla^2 H)_{tt}^{(m)} = \sum_{p=1}^{N-1} \frac{-(1 - \cos(2\pi mp/N) \cos(2\pi p/N)) + 2(1 - \cos(2\pi mp/N)) \cos^2(\pi p/N)}{4 \sin^2(\pi p/N)}. \quad (15)$$

The bracketed numerator simplifies via $2 \cos^2(\pi p/N) = 1 + \cos(2\pi p/N)$:

$$\begin{aligned} & - (1 - \cos(2\pi mp/N) \cos(2\pi p/N)) + (1 - \cos(2\pi mp/N))(1 + \cos(2\pi p/N)) \\ & = \cos(2\pi p/N) - \cos(2\pi mp/N). \end{aligned}$$

(The $\cos(2\pi mp/N) \cos(2\pi p/N)$ cross terms cancel, $-1 + 1 = 0$, and the remaining terms are as stated.)

Thus

$$(\nabla^2 H)_{tt}^{(m)} = \sum_{p=1}^{N-1} \frac{\cos(2\pi p/N) - \cos(2\pi mp/N)}{4 \sin^2(\pi p/N)} = C(1) - C(m), \quad (16)$$

where $C(k)$ is as in Corollary 1. Substituting,

$$(\nabla^2 H)_{tt}^{(m)} = \frac{m(N-m)}{2} - \frac{N-1}{2} = \frac{m(N-m) - (N-1)}{2}. \quad (17)$$

For $m \in \{1, \dots, N-1\}$, $m(N-m) \geq N-1$ with equality exactly at $m \in \{1, N-1\}$, so $(\nabla^2 H)_{tt}^{(m)} \geq 0$ with equality at the translation modes (consistent with the translation Goldstone structure: modes $m = 1$ and $m = N-1$ generate infinitesimal rigid translations, which lie in the kernel of $\nabla^2 H$).

5 Off-diagonal vanishing and trace identity

Lemma 4 (Off-diagonal vanishing). *For every $N \geq 2$ and every $m \in \{1, \dots, N-1\}$ the radial-tangential cross entry of $\nabla^2 H$ in the mode- m normalized Fourier basis vanishes:*

$$(\nabla^2 H)_{rt}^{(m)} = 0.$$

Proof. Under the combined perturbation $\delta z_k = z_k \alpha_k (a + ib)$ with $\alpha_k = \cos(2\pi mk/N)$, write $A = z_j - z_k$ and $B = \alpha_j z_j - \alpha_k z_k$. Then $z_j(a, b) - z_k(a, b) = A + (a + ib)B$, so

$$|z_j(a, b) - z_k(a, b)|^2 = |A|^2 + 2a \operatorname{Re}(A\bar{B}) + 2b \operatorname{Im}(A\bar{B}) + (a^2 + b^2) |B|^2.$$

Crucially, there is *no* ab term in $|v(a, b)|^2$ itself. The ab term in $-\frac{1}{2} \ln |v(a, b)|^2$ therefore comes entirely from the cross product of the two linear terms in the expansion $-\frac{1}{2} \ln(1+x) = -x/2 + x^2/4 - \dots$, namely

$$\text{coeff of } ab \text{ in } -\frac{1}{2} \ln |v(a, b)|^2 = \frac{P_{jk} R_{jk}}{2 U_{jk}^2},$$

with $P_{jk} = 2 \operatorname{Re}(A\bar{B}) = 2(\alpha_j + \alpha_k)(1 - \cos(\theta_j - \theta_k))$ and $R_{jk} = 2 \operatorname{Im}(A\bar{B}) = 2(\alpha_j - \alpha_k) \sin(\theta_j - \theta_k)$. Multiplying,

$$P_{jk}R_{jk} = 4(\alpha_j + \alpha_k)(\alpha_j - \alpha_k)(1 - \cos(\theta_j - \theta_k)) \sin(\theta_j - \theta_k) = 4(\alpha_j^2 - \alpha_k^2)(1 - \cos(\theta_j - \theta_k)) \sin(\theta_j - \theta_k).$$

Using $1 - \cos(\theta_j - \theta_k) = U_{jk}/2$,

$$\frac{P_{jk}R_{jk}}{2U_{jk}^2} = \frac{2(\alpha_j^2 - \alpha_k^2) \sin(\theta_j - \theta_k)}{U_{jk}}.$$

Summing over pairs with $p = k - j$ and fixing p , the dependence on j is only through $\alpha_j^2 - \alpha_{j+p}^2$. But $\sum_{j=0}^{N-1} \alpha_j^2 = N/2 = \sum_{j=0}^{N-1} \alpha_{j+p}^2$ (the second sum is a cyclic shift of the first), so

$$\sum_{j=0}^{N-1} (\alpha_j^2 - \alpha_{j+p}^2) = 0 \quad \text{for every } p \in \{1, \dots, N-1\}.$$

Therefore the ab coefficient in $H(a, b)$ vanishes, which (after unit normalization of $\hat{e}_r, \hat{e}_t$) gives $(\nabla^2 H)_{rt}^{(m)} = 0$. $\square$

Lemma 5 (Trace identity). $\lambda_m^+ + \lambda_m^- = N - 1$ for every $m \in \{1, \dots, N-1\}$.

Proof. Direct from (19): $[(N-1) - m(N-m)/2] + [m(N-m)/2] = N-1$. Equivalently, $\lambda_m^+ + \lambda_m^- = (\nabla^2 H)_{rr}^{(m)} + (\nabla^2 H)_{tt}^{(m)} + 2 \cdot (N-1)/2$, and the two unshifted sums telescope via (13) and (17): $(N-1 - m(N-m))/2 + (m(N-m) - (N-1))/2 = 0$, leaving $N-1$. $\square$

6 Lagrange shift and eigenvalues

The Lagrangian is $\mathcal{L} = H - \mu_L L$ with angular impulse $L = \sum_k |z_k|^2$ and $\mu_L = -(N-1)/4$ at the unit-ring equilibrium. Since $\nabla^2 L = 2I$, the Lagrangian Hessian is

$$\nabla^2 \mathcal{L} = \nabla^2 H - 2\mu_L I = \nabla^2 H + \frac{N-1}{2} I. \quad (18)$$

Adding $(N-1)/2$ to each of (13) and (17) yields

$$\boxed{\lambda_m^+ = \frac{N-1-m(N-m)}{2} + \frac{N-1}{2} = (N-1) - \frac{m(N-m)}{2}, \quad \lambda_m^- = \frac{m(N-m) - (N-1)}{2} + \frac{N-1}{2} = \frac{m(N-m)}{2}} \quad (19)$$

The trace identity $\lambda_m^+ + \lambda_m^- = N-1$ falls out automatically — it is not an independent check but a consequence of the $(N-1)/2$ shift doubling across the two diagonal entries.

Remark 1 (The case $m = N/2$ for even N). When N is even and $m = N/2$, set $\alpha_k = \cos(\pi k) = (-1)^k$. The mode vector $\hat{e}_r$ has squared norm $\|\hat{e}_r\|^2 = \sum_k \alpha_k^2 = N$ (because $\alpha_k^2 = 1$ for every k , not averaged to $1/2$). The key Fourier sums change correspondingly to

$$\sum_{j=0}^{N-1} \alpha_j \alpha_{j+p} = \sum_j (-1)^{2j+p} = N(-1)^p = N \cos(2\pi \cdot \frac{N}{2} \cdot p/N),$$

which is exactly twice the generic expression $(N/2) \cos(2\pi mp/N)$ evaluated at $m = N/2$. Consequently $\sum_j (\alpha_j + \alpha_{j+p})^2 = 2N(1 + \cos(\pi p))$ and $\sum_j Q_{j,j+p} = 2N(1 - \cos(\pi p) \cos(2\pi p/N))$ —

each double the generic value. Substituting into the normalized Hessian entry with $\|\hat{e}_r\|^2 = N$ (so the prefactor is $2/N$, not $4/N$), the two factors of 2 cancel and the same algebra goes through. Explicitly,

$$(\nabla^2 H)_{rr}^{(N/2)} = \sum_p \frac{1+\cos(\pi p)}{2} - \sum_p \frac{1-\cos(\pi p)\cos(2\pi p/N)}{4\sin^2(\pi p/N)} = \frac{N-1-(N/2)(N/2)}{2} = \frac{N-1-N^2/4}{2},$$

matching $(N-1-m(N-m))/2$ at $m = N/2$. The tangential-tangential entry follows identically. The final eigenvalue formula (19) is thus uniform in $m \in \{1, \dots, N-1\}$, including $m = N/2$.

7 Numerical verification

The derivation above is self-contained; the script `numerical.check.py` in this directory corroborates each step with machine-checked identity verification.

Primary verification — symbolic to 30 decimal digits. Every intermediate identity used in the derivation is verified via `sympy` arbitrary-precision evaluation for $N \in \{3, \dots, 10\}$ and $m \in \{1, \dots, N-1\}$: the Havelock identity (5), the cosecant-squared sum (7), the radial-radial and tangential-tangential unshifted expressions (13) and (17), and the α^2 -translation identity in the off-diagonal proof. Each is confirmed to differ from its closed form by less than 10^{-25} .

Corroborating sanity check — finite-difference oracle. As an independent cross-check that the block-level algebra is correctly assembled, the script diagonalizes the full $2N \times 2N$ Hessian $\nabla^2 H$ at the unit ring (via central differences), projects into the normalized mode- m (radial, tangential) basis, applies the Lagrange shift, and compares to (19). For $N \in \{3, \dots, 12\}$ and all $m \in \{1, \dots, N-1\}$, the observed eigenvalues match to 10^{-2} (finite-difference precision). The three canonical cases $(N, m) \in \{(5, 2), (7, 3), (11, 5)\}$ match to machine precision.